

## SEPT-ILES, QC, Canada

WMO: 718110

Lat: 50.2181N

Lon: 66.2414W

Elev: 180

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.2      | -13.3 | -29.7                         | 1.0 | -17.0 | -25.0 | 1.4 | -12.0 | 29.3                  | 21.0 | 24.9 | 18.5 | 6.3                   | 300  | 0.712 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.9                   | 73.4            | 61.7 | 70.6 | 60.3 | 68.2 | 59.4 | 64.6                | 69.6 | 63.0 | 67.3 | 61.4 | 65.6 | 10.1                 | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.7                            | 85.7 | 66.6 | 61.1 | 81.1 | 64.9 | 59.7 | 76.9 | 63.5 | 29.9          | 69.8 | 28.6 | 67.5 | 27.5 | 65.6 | 70.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 23.5              | 20.6 | 18.2 | -27.4                      | 80.2 | 5.0                | 2.9 | -31.0                                              | 82.3 | -33.9      | 84.0 | -36.7      | 85.6 | -40.3      | 87.7 |  |  |
|                   |      |      | -27.7                      | 67.9 | 4.8                | 1.7 | -31.2                                              | 69.2 | -34.0      | 70.2 | -36.8      | 71.1 | -40.3      | 72.4 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 35.0   | 7.1   | 9.2   | 19.6  | 32.2  | 43.1  | 53.4  | 59.7  | 58.8  | 50.4  | 40.2  | 28.6  | 15.5  |
|                                                                        | DBStd       | 19.89  | 11.98 | 11.77 | 10.74 | 6.86  | 5.56  | 5.73  | 4.28  | 4.57  | 5.72  | 5.82  | 8.54  | 10.99 |
|                                                                        | HDD50       | 6286   | 1329  | 1141  | 942   | 535   | 225   | 29    | 0     | 2     | 63    | 309   | 642   | 1069  |
|                                                                        | HDD65       | 10980  | 1794  | 1561  | 1407  | 985   | 679   | 351   | 172   | 197   | 438   | 770   | 1092  | 1534  |
|                                                                        | CDD50       | 798    | 0     | 0     | 0     | 0     | 12    | 130   | 301   | 276   | 75    | 4     | 0     | 0     |
|                                                                        | CDD65       | 15     | 0     | 0     | 0     | 0     | 0     | 1     | 8     | 6     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 69     | 0     | 0     | 0     | 0     | 1     | 13    | 32    | 21    | 2     | 0     | 0     | 0     |
|                                                                        | CDH80       | 4      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 7.9    | 8.5   | 8.6   | 8.9   | 9.1   | 8.0   | 7.8   | 6.9   | 6.4   | 6.8   | 7.8   | 8.0   | 8.5   |
|                                                                        | PrecAvg     | 46.2   | 3.6   | 3.0   | 3.4   | 3.6   | 3.7   | 3.8   | 4.0   | 3.9   | 4.5   | 4.3   | 4.3   | 4.3   |
| (15) Precipitation                                                     | PrecMax     | 60.8   | 8.1   | 7.4   | 8.6   | 10.4  | 7.1   | 7.8   | 7.1   | 9.0   | 9.2   | 8.5   | 9.4   | 9.2   |
|                                                                        | PrecMin     | 35.7   | 0.0   | 0.3   | 0.8   | 0.4   | 0.7   | 1.3   | 1.6   | 1.6   | 1.4   | 0.8   | 1.5   | 1.5   |
|                                                                        | PrecStd     | 6.9    | 1.6   | 1.7   | 1.7   | 2.2   | 1.7   | 1.5   | 1.3   | 1.8   | 1.7   | 1.8   | 1.7   | 1.7   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.7  | 39.3  | 44.0  | 53.0  | 67.9  | 75.6  | 78.0  | 76.7  | 71.1  | 61.4  | 50.0  | 41.1  |
|                                                                        |             | MCWB   | 34.6  | 37.0  | 37.1  | 42.0  | 53.0  | 59.7  | 63.0  | 63.7  | 60.9  | 53.2  | 47.7  | 39.5  |
|                                                                        | 2%          | DB     | 33.1  | 34.6  | 39.2  | 47.8  | 61.0  | 70.6  | 73.5  | 72.8  | 66.2  | 55.2  | 45.0  | 37.6  |
|                                                                        |             | MCWB   | 31.8  | 32.4  | 34.5  | 39.5  | 50.2  | 58.4  | 62.1  | 62.1  | 57.2  | 49.1  | 42.3  | 35.8  |
|                                                                        | 5%          | DB     | 29.7  | 31.5  | 36.0  | 44.2  | 56.6  | 67.2  | 70.7  | 69.8  | 62.7  | 51.9  | 42.5  | 34.6  |
|                                                                        |             | MCWB   | 28.4  | 29.6  | 32.2  | 37.5  | 47.4  | 56.7  | 60.9  | 60.7  | 55.7  | 47.4  | 40.3  | 33.1  |
|                                                                        | 10%         | DB     | 25.9  | 27.4  | 33.6  | 41.3  | 53.2  | 63.8  | 68.2  | 67.3  | 60.0  | 49.2  | 40.0  | 32.0  |
|                                                                        |             | MCWB   | 24.3  | 25.5  | 30.7  | 36.1  | 46.1  | 55.0  | 60.0  | 60.1  | 54.5  | 45.7  | 37.9  | 30.8  |
|                                                                        | 0.4%        | WB     | 35.1  | 37.5  | 38.7  | 43.6  | 55.8  | 64.4  | 67.1  | 66.5  | 63.0  | 54.6  | 48.3  | 40.3  |
|                                                                        |             | MCDB   | 36.6  | 38.8  | 42.7  | 49.3  | 64.5  | 69.9  | 73.2  | 72.3  | 68.3  | 58.0  | 49.3  | 40.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 32.0  | 33.0  | 35.2  | 41.0  | 51.6  | 60.9  | 64.8  | 64.5  | 59.6  | 51.1  | 43.3  | 36.2  |
|                                                                        |             | MCDB   | 33.1  | 34.1  | 37.9  | 46.1  | 58.8  | 66.9  | 69.8  | 69.4  | 63.4  | 54.1  | 44.2  | 37.2  |
|                                                                        | 5%          | WB     | 28.6  | 29.5  | 33.0  | 38.9  | 49.2  | 58.6  | 63.2  | 63.0  | 57.4  | 48.5  | 40.6  | 33.2  |
|                                                                        |             | MCDB   | 29.7  | 31.4  | 35.2  | 43.0  | 54.7  | 64.5  | 67.7  | 67.1  | 60.8  | 50.7  | 42.0  | 34.4  |
|                                                                        | 10%         | WB     | 24.3  | 25.6  | 31.1  | 36.9  | 46.9  | 56.4  | 61.6  | 61.5  | 55.4  | 46.4  | 38.1  | 30.9  |
|                                                                        |             | MCDB   | 25.7  | 27.3  | 33.3  | 40.5  | 51.9  | 61.6  | 65.8  | 65.4  | 58.9  | 48.4  | 39.7  | 32.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.9  | 18.8  | 17.0  | 13.3  | 15.2  | 15.9  | 14.9  | 15.2  | 15.7  | 13.3  | 12.4  | 15.0  |
|                                                                        |             | MCDBR  | 17.6  | 14.5  | 14.5  | 17.4  | 22.5  | 22.8  | 19.8  | 19.4  | 19.6  | 17.9  | 11.6  | 12.3  |
|                                                                        |             | MCWBR  | 17.4  | 13.5  | 11.4  | 11.4  | 14.3  | 12.9  | 11.2  | 11.3  | 13.0  | 12.8  | 10.5  | 12.3  |
|                                                                        | 5% WB       | MCDBR  | 17.7  | 13.7  | 13.1  | 14.6  | 19.8  | 18.9  | 15.6  | 15.4  | 16.6  | 14.2  | 10.9  | 12.4  |
|                                                                        |             | MCWBR  | 17.7  | 13.1  | 11.3  | 10.5  | 13.5  | 12.2  | 10.3  | 10.2  | 12.5  | 11.6  | 10.8  | 12.8  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.262 | 0.268 | 0.293 | 0.342 | 0.365 | 0.391 | 0.425 | 0.386 | 0.363 | 0.327 | 0.294 | 0.267 |
|                                                                        | taud        |        | 2.363 | 2.323 | 2.323 | 2.308 | 2.393 | 2.369 | 2.306 | 2.418 | 2.454 | 2.525 | 2.504 | 2.393 |
|                                                                        | Ebn at Noon |        | 252   | 280   | 291   | 286   | 283   | 275   | 263   | 269   | 264   | 256   | 239   | 232   |
|                                                                        | Edh at Noon |        | 20    | 27    | 32    | 37    | 36    | 37    | 39    | 33    | 29    | 23    | 18    | 17    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 353   | 674   | 1091  | 1435  | 1654  | 1746  | 1636  | 1386  | 1042  | 628   | 333   | 257   |
|                                                                        | RadStd      |        | 40    | 61    | 88    | 118   | 104   | 148   | 160   | 85    | 73    | 52    | 31    | 22    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | +0.83 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (10 neighbors) | N/A   | N/A     | +2.25  | N/A     | +0.47 | +0.61 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page